Appendix

1. Supplementary Material Roadmap

This supplementary material collects the extra evidence and reproducibility details that are too long for the main paper. It is organized so that a reviewer can check the claims in order. We first provide additional SCM diagnostics, then describe the TCF construction and verification protocol, then give the experimental and baseline settings. We next list the naturalness-oriented variants, the complete model pool, the full per-suspect results, and the remaining naturalness and distillation diagnostics.

2. Additional SCM Diagnostics

This section gives additional model-wise diagnostics for the SCM-based analysis. The figures are used only as supporting evidence: they show how SCM changes during source-side optimization, how derived models preserve the target behavior, how independent models differ, and how these effects translate into target-answer separation.

2.1. SCM Control across Additional Source Families. Figure 13 checks whether the source-side optimization behaves consistently on the other source models. Across Qwen3 Instruct, Llama3, and Mistral, the target loss generally decreases during GCG optimization, while the mean SCM increases. This supports the use of target-loss minimization as a practical way to strengthen the source-side counterfactual transition.

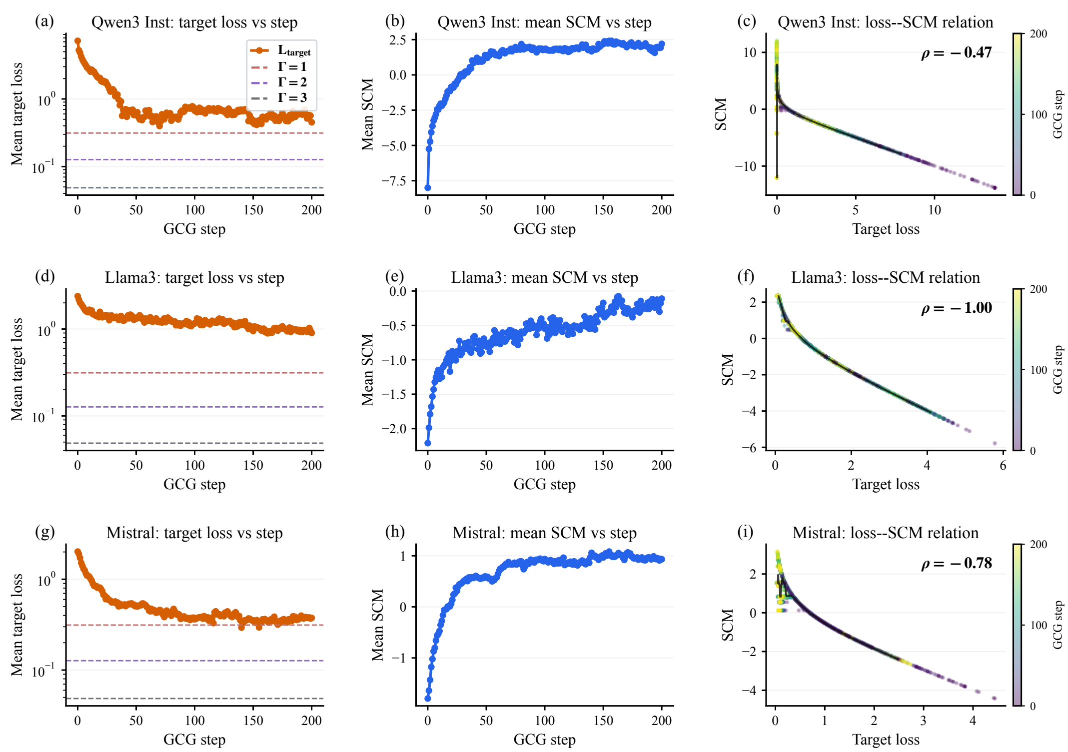

Figure 13: Additional SCM-control diagnostics on three source families. GCG optimization reduces the source target loss, increases the mean SCM, and produces a clear negative relation between target loss and SCM.

2.2. Derived-Model Preservation Diagnostics. Figure 14 checks whether derived models preserve the source-induced target behavior on the additional source families. In each row, the left plot shows how much weakening occurs when moving from the source model to its derivatives, and the right plot shows the corresponding lower-bound trend. Smaller weakening budgets and higher lower-bound curves indicate stronger preservation by derived models.

2.3. Independent-Model Non-Transfer Diagnostics. Figure 15 checks whether the same optimized prompts also transfer to unrelated models. The plots show that independent models require larger slack budgets, meaning that they do not preserve the source-side target preference as consistently as derived models. This supports the intended separation between source-family models and unrelated models.

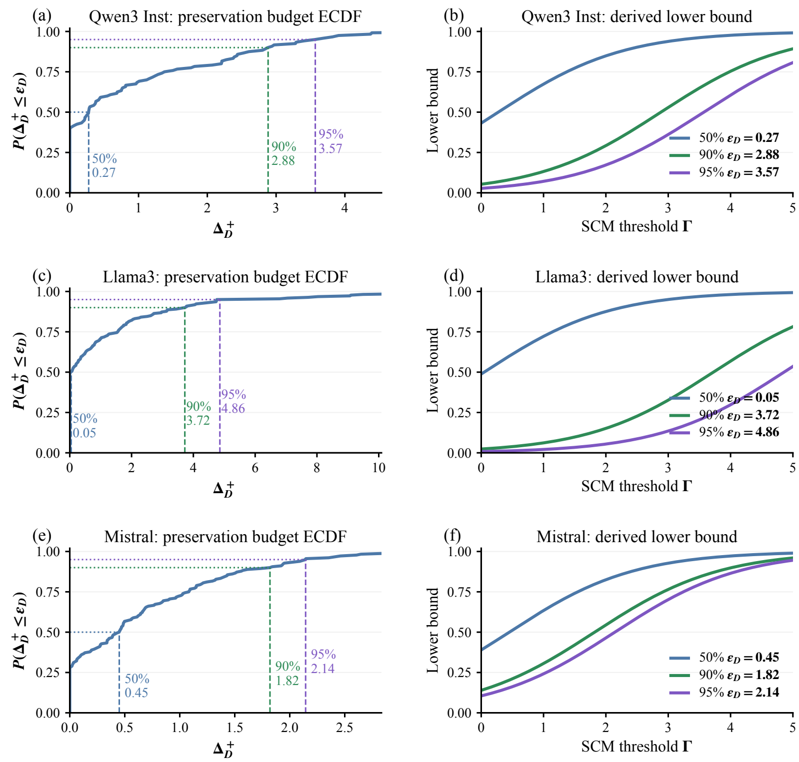

Figure 14: Additional derived-model preservation diagnostics. For each row, the left plot shows the empirical preservation budget ϵ_D from $\Pr(\Delta_D^+ \leq \epsilon_D)$ over derived models, and the right plot shows the lower-bound trend $\sigma(\Gamma - \epsilon_D)$.

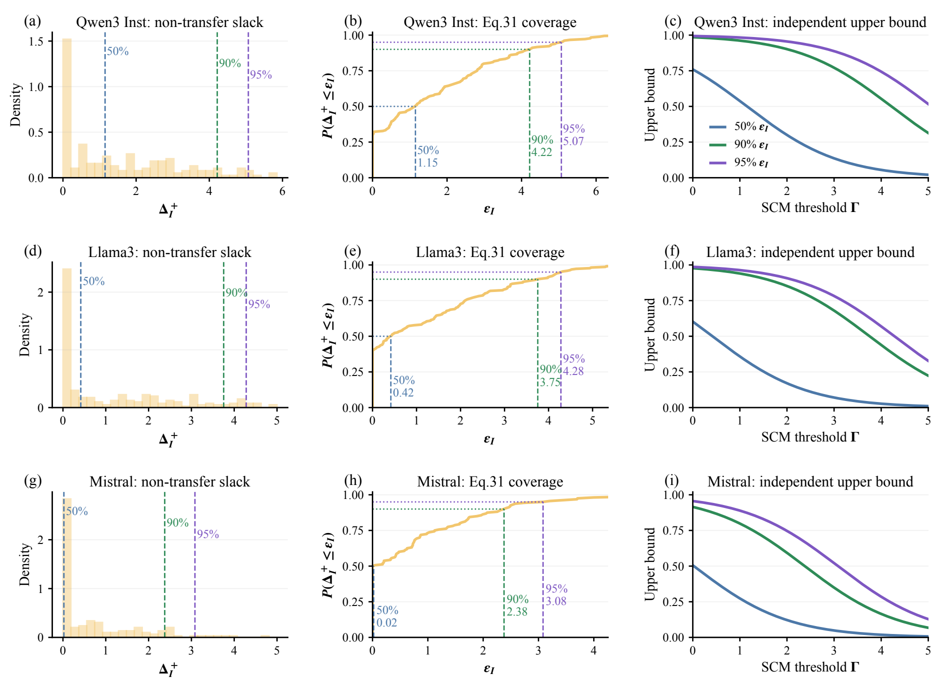

Figure 15: Additional independent-model non-transfer diagnostics. For each row, the left plot shows the non-transfer slack distribution, the middle plot shows the empirical coverage $\Pr(\Delta_I^+ \leq \epsilon_I)$, and the right plot shows the corresponding independent-model upper-bound trend $\sigma(-\Gamma + \epsilon_I)$.

2.4. SCM-Induced Separability Diagnostics. Figure 16 connects the previous diagnostics to the final target-answer gap. The empirical bucket plots show that higher-SCM fingerprints usually increase the gap between derived and independent models, while the bound plots show the same trend under the measured budgets. The main message is that SCM helps select stronger fingerprints, but over-optimization is still not the goal.

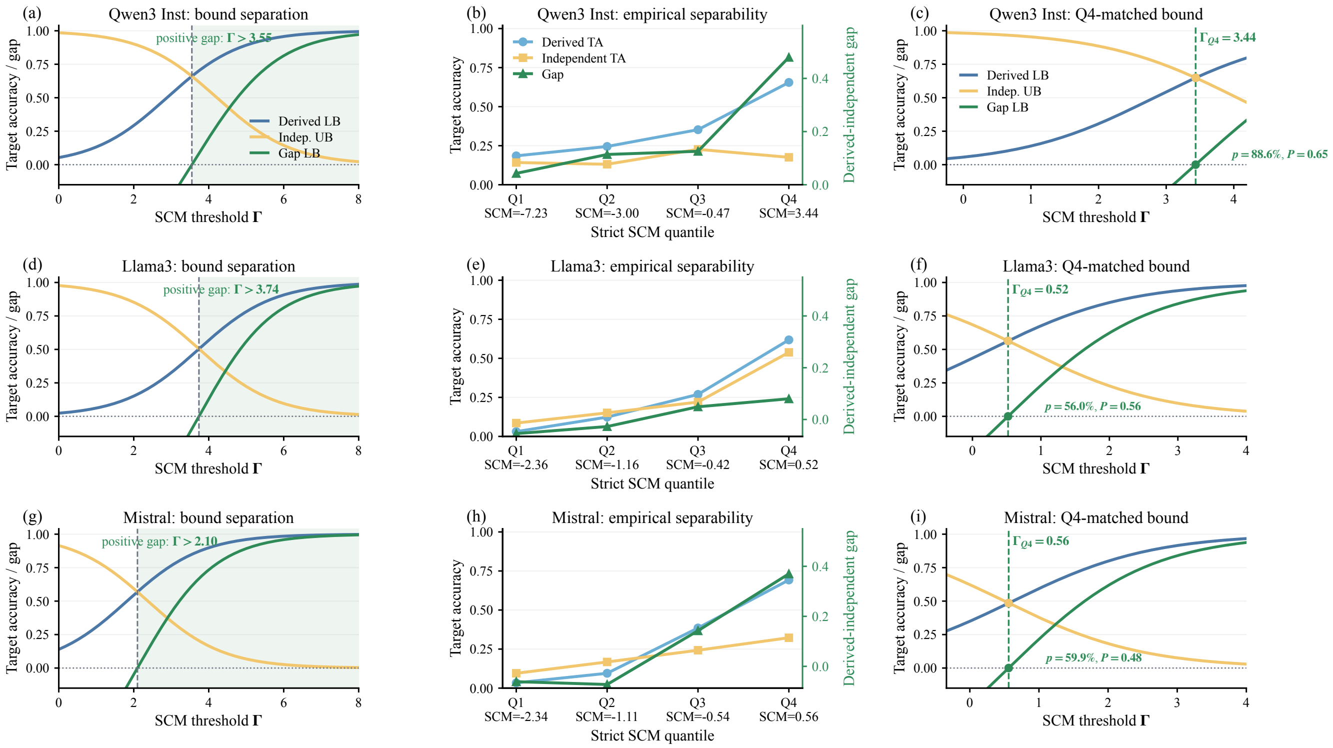

Figure 16: Additional SCM-induced separability diagnostics. The left column shows conservative bound separation using empirical budgets; the middle column shows empirical target-accuracy trends across SCM buckets; the right column shows the bound trend after matching the high-SCM bucket. These results support the same reading as the main analysis: stronger SCM usually improves the derived–independent gap, but the best operating point is not obtained by over-optimizing indefinitely.

How to read these figures. Figures 13–16 follow the same evidence chain. Source-side optimization increases SCM. Derived models tend to keep the target preference more often than independent models. As a result, higher-SCM fingerprints usually give a larger target-answer gap between the two groups.

2.5. Logit-Diagnostic Breakdowns. The compact diagnostics above average over many model–fingerprint pairs. Figure 17 shows where the derived-preservation and independent non-transfer effects come from. Panels (a)–(d) report the positive weakening drift $\Delta_D^+ = \max(0, L_{M_0}(p^u, t) - L_M(p^u, t))$ for the four protected source families. Smaller drift means that a derived model keeps the source model's prefixed target preference more strongly. Panel (e) compares the prefix-induced target-log-odds lift on source and independent models.

Observation. Figure 17 explains the aggregate trends used in the SCM diagnostics. Source-family derivatives usually need a smaller weakening budget than unrelated models, while the optimized prefix produces a stronger target-log-odds lift on the source side than on independent models.

3. Additional Theory

This section gives additional theory explanations that support the main analysis and the concise paper appendix.

3.1. Interpretation of Local Behavioral Closeness. The concise interpretation is provided in the paper appendix. For completeness, we retain the longer heuristic details here. By construction, the source endpoints satisfy

$$m_{M_0}(p^0(x), t) \leq -\Gamma, \qquad m_{M_0}(p^u(x), t) \geq +\Gamma, \tag{66}$$

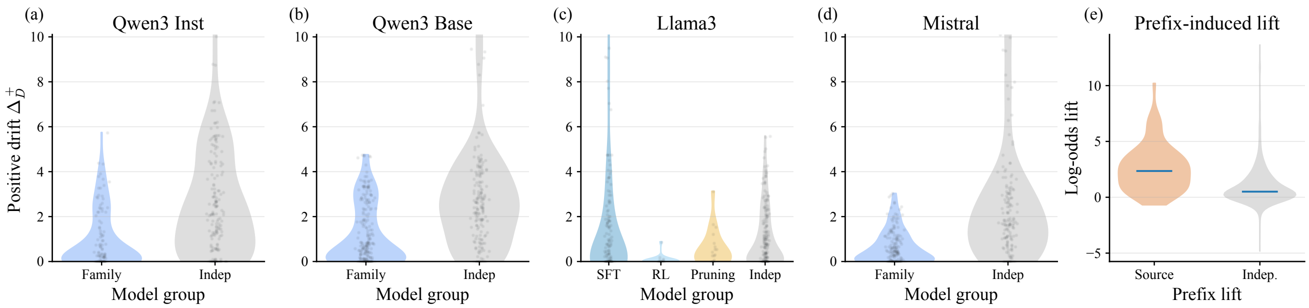

Figure 17: Source-family breakdown of the logit diagnostics used for the derived-preservation and independent non-transfer analysis. Panels (a)–(d) report the positive weakening drift Δ_D^+ by source family and model group. Panel (e) compares the prefix-induced target-log-odds lift between the source and independent models.

under target selection and after the perturbation reaches the source margin. For common derivation operations, a bounded parameter change such as

$$\|\theta_D - \theta_0\| \le \rho \tag{67}$$

combined with a locally Lipschitz logit map gives the heuristic relation

$$\beta \lesssim L\rho. \tag{68}$$

This is only an interpretation of the local-closeness premise; the verification statistic still relies on the measured hard-label target behavior.

3.2. Derived-Model Preservation Details. Closeness of the target behavior implies that, for the fingerprints covered by the premise, a derived model should retain the source model's perturbed target preference up to some weakening. We keep the bound conditional on a fingerprint-wise weakening budget and use the empirical cumulative distribution function (ECDF) to estimate how often this budget is small.

In the Qwen3-1.7B Base diagnostic, we compute the weakening drift Δ_D^+ between the source model and its derivatives. The ECDF reports the fraction of derived model–fingerprint pairs whose weakening is at most a chosen budget ϵ_D. A smaller budget means stronger preservation. Substituting these budgets into the lower bound shows the same trend: higher SCM gives a stronger conditional lower bound, while higher-coverage budgets give more conservative bounds.

3.3. Independent-Model Non-Transfer Details. The independent side of the analysis states that an independent model does not stay uniformly close to the protected source model at the perturbed input. The perturbation u is optimized only against M_0, so it follows the local behavior of M_0. On a model whose local behavior is not aligned with M_0, the same perturbation should produce only a weak generic effect.

This is consistent with the known asymmetry of targeted adversarial transfer: when a perturbation must push the output to one specific target, it usually transfers less reliably across independently trained models than untargeted perturbations. A simple first-order reading is

$$m_{M_I}(p^u, t) \approx m_{M_I}(p^0, t) + \langle \nabla_u m_{M_I}, u \rangle, \tag{69}$$

where the induced gain depends on the cross-model gradient alignment

$$\kappa = \cos(\nabla_u m_{M_0}, \nabla_u m_{M_I}). \tag{70}$$

For independent models, this alignment is expected to be small. Therefore, the claim is not that independent target accuracy must decrease monotonically with SCM; rather, for a fixed amount of generic-transfer slack, larger SCM gives a tighter upper bound on independent-model target accuracy.

The non-transfer argument also needs the target to be unlikely for the independent model's own clean prompt:

$$m_{M_I}(p^0(x), t) \le 0. \tag{71}$$

Under this anchor, the desired non-transfer reading is

$$L_{M_I}(p^u(x), t) \approx L_{M_I}(p^0(x), t). \tag{72}$$

If an unrelated model already prefers the same target for content reasons, that example can become a residual false positive. This is why target diversity and counterfactual target selection are important: they reduce the chance that many independent models naturally prefer the recorded targets.

3.4. Separability Details and Feasible Margin Band. For fixed derived-model and independent-model transfer budgets, increasing SCM improves the gap between the derived-model lower bound and the independent-model upper bound. Formally, for fixed ϵ_D and ϵ_I,

$$\frac{\partial g}{\partial \Gamma} = \sigma'(\Gamma - \epsilon_D) + \sigma'(-\Gamma + \epsilon_I) > 0, \tag{73}$$

where $\sigma'(z) = \sigma(z)(1 - \sigma(z))$. Thus, larger SCM increases the fixed-budget gap envelope. The gain saturates as $\Gamma \to \infty$ because both sigmoid derivatives vanish in the tails.

However, this does not mean that larger SCM is always better. Reaching a very large source margin usually requires a stronger perturbation u, which can also enlarge the generic-transfer term

$$\langle \nabla_u m_{M_I}, u \rangle \tag{74}$$

and hence ϵ_I. The two effects define a feasible margin band

$$[\Gamma_{\mathrm{lo}}, \Gamma_{\mathrm{hi}}]. \tag{75}$$

The lower edge is approximately

$$\Gamma_{\mathrm{lo}} \approx \beta + c_D, \tag{76}$$

which is large enough for derived models to survive their weakening and still flip. The upper edge Γ_{hi} is small enough that the perturbation has not yet become generic enough to transfer strongly to independent models.

This gives the feasible-margin interpretation used in the paper. Inside the band, the expected gap $g(\Gamma)$ stays positive; pushing Γ past Γ_{hi} can inflate ϵ_I faster than the saturating derived gain, so the empirical gap can shrink. In the Qwen3-Base diagnostic, using the 90% empirical budgets gives the conservative sufficient condition

$$\Gamma > \frac{\epsilon_D + \epsilon_I}{2} \approx 3.48. \tag{77}$$

This condition is sufficient, not necessary. In practice, the construction does not compute the band edges explicitly; it uses a source-side stopping rule and stops once the target condition is reached.

4. TCF Construction and Verification Details

This section provides the full TCF construction and verification details.

4.1. Adaptive SCM Progress-to-Cap Selection. A fixed GCG step count is not equally suitable for every source model and every question. Some candidates reach a good source-side counterfactual transition early, while others need more updates. We therefore use the SCM trajectory to choose a checkpoint for each candidate.

For each candidate fingerprint i and each recorded GCG checkpoint $\kappa \in \mathcal{K} = \{\kappa_1 < \cdots < \kappa_m\}$, let $u_i^{(\kappa)}$ be the prefix at checkpoint κ and define

$$\Gamma_i(\kappa) = \Gamma(x_i, u_i^{(\kappa)}, t_i). \tag{78}$$

Because the clean side of SCM is fixed once the target is selected, each candidate has a source-side SCM cap

$$c_i = -L_{M_0}(p^0(x_i), t_i). \tag{79}$$

We normalize each candidate's progress as

$$r_i(\kappa) = \mathrm{clip}_{[0,1]}\left(\frac{\Gamma_i(\kappa) - \Gamma_i(\kappa_1)}{c_i - \Gamma_i(\kappa_1) + \epsilon_{\mathrm{prog}}}\right), \tag{80}$$

where $\epsilon_{\mathrm{prog}} > 0$ only avoids division by zero. The value $r_i(\kappa)$ tells us how much of the available SCM improvement has been achieved by checkpoint κ.

We estimate a source-specific progress level from a source-only discovery pool $\mathcal{C}_{\mathrm{disc}}$:

$$\bar{r}(\kappa_j) = \frac{1}{|\mathcal{C}_{\mathrm{disc}}|} \sum_{i \in \mathcal{C}_{\mathrm{disc}}} r_i(\kappa_j). \tag{81}$$

Let $\widetilde{\kappa}_j = (\kappa_j - \kappa_1)/(\kappa_m - \kappa_1)$ be the normalized checkpoint index. We choose the elbow point

$$j^\star = \arg\max_{1 \le j \le m} \left[\bar{r}(\kappa_j) - \widetilde{\kappa}_j\right], \qquad \rho^\star = \bar{r}(\kappa_{j^\star}). \tag{82}$$

This step uses only the protected source model. It replaces a fixed iteration horizon with a source-specific progress level.

For each candidate, we keep the earliest checkpoint that reaches this progress level:

$$h_i = \min\{\kappa_j \in \mathcal{K} : r_i(\kappa_j) \geq \rho^\star\}. \tag{83}$$

If no checkpoint reaches $\rho^\star$, we use the checkpoint with the largest SCM. The retained prefix is $u_i^\star = u_i^{(h_i)}$, and the source-side SCM gain used for sample ranking is

$$S_i = \Gamma_i(h_i) - \Gamma_i(\kappa_1). \tag{84}$$

Candidates are then filtered by the SCM threshold, the source-stability check, and the diversity constraints on questions, clean answers, and target answers. No derived-model, independent-model, or suspect-model statistic is used in this selection rule.

4.2. SCM-Guided Fingerprint Construction. Algorithm 1 summarizes how we construct candidate fingerprints on the source model. The construction first selects a clean-unlikely target, then optimizes a GCG prefix toward that target, and finally keeps only candidates that pass the SCM and stability checks.

Algorithm 1 SCM-guided GCG fingerprint construction

Require: Source model M_0; question pool $\mathcal{D}$; constrained answer space $\mathcal{A}$; parser Parse; desired fingerprint number n; SCM threshold $\Gamma_{\min}$; prefix length ℓ_u; GCG budget $T_{\max}$; stability threshold $s_{\min}$.
Ensure: Fingerprint set $\mathcal{F}$.

```
F ← ∅
for each question x ∈ D do
    Construct clean prompt p^0(x) and estimate q_0(a | p^0(x)) for all a ∈ A.
    y ← arg max_{a∈A} q_0(a | p^0(x)).
    T ← {a ∈ A \ {y} : −L_{M_0}(p^0(x), a) ≥ Γ_min}.
    if T = ∅ then
        continue
    end if
    t ← arg min_{a∈T} q_0(a | p^0(x)).
    Initialize a prefix u with length ℓ_u.
    for j = 1 to T_max do
        Use one GCG update to reduce the source target loss for target t.
        if L_GCG(u; x, t) ≤ log(1 + e^{−Γ_min}) then
            break
        end if
    end for
    Compute Γ(x, u, t) using the SCM definition.
    if Γ(x, u, t) < Γ_min then
        continue
    end if
    Estimate source stability ŝ_0(x, u, t) by repeated source queries.
    if ŝ_0(x, u, t) < s_min then
        continue
    end if
    Add (p^u(x), y, t, u, Γ(x, u, t)) to F.
    if |F| = n then
        break
    end if
end for
return F
```

4.3. Hard-Label Ownership Verification. Algorithm 2 gives the online verification rule. The verifier only queries the suspect model, parses the final response into the legal answer set, and counts how often the parsed label matches the recorded target.

Algorithm 2 Hard-label ownership verification

Require: Fingerprint set $\mathcal{F}$; suspect black-box model M; parser Parse; threshold ζ.
Ensure: Ownership decision.

$S \leftarrow 0$
for each fingerprint $(p_i, y_i, t_i, u_i, \Gamma_i) \in \mathcal{F}$ **do**
  Query M with p_i and obtain raw response r_i.
  $\hat{a}_i \leftarrow \mathrm{Parse}(r_i)$.
  if $\hat{a}_i = t_i$ **then**
    $S \leftarrow S + 1$
  end if
end for
$\mathrm{TA}(M; \mathcal{F}) \leftarrow S/|\mathcal{F}|$.
if $\mathrm{TA}(M; \mathcal{F}) \geq \zeta$ **then**
  return "derived"
else
  return "independent"
end if

4.4. Thresholds, Parsing, and Invalid Outputs. Constrained answer space. The framework supports any finite legal output space. In our experiments, we use multiple-choice labels $\{A, B, C, D\}$ because they are easy to parse and make verification strictly hard-label.

SCM threshold. The main construction parameter is $\Gamma_{\min}$. A smaller value keeps more candidates; a larger value keeps only stronger candidates but may reduce diversity. In practice, the threshold should be chosen from source-side feasibility, not from suspect-model results.

GCG stopping rule. GCG stops when the source target loss is at most $\log(1 + e^{-\Gamma_{\min}})$ or when the maximum iteration budget is exhausted. This aligns optimization with the SCM acceptance rule.

Stability threshold. We use a repeated-query source-stability check with threshold $s_{\min} \in [0.8, 0.95]$. A higher value is preferred when the API uses stochastic decoding.

Invalid outputs. The prompt asks for one valid label only. Any output that cannot be parsed into the legal answer set is counted as a failure. This avoids subjective semantic matching of free-form text.

5. Experimental Protocol and Baseline Configuration

This section gives the experimental details needed to reproduce the reported verification results. It first defines the prompt examples used in this supplement, then gives the fingerprint dataset and parameters, and finally lists the baseline settings and representative baseline artifacts.

Table 13 summarizes the prompt examples and baseline artifacts used in this supplementary material. The full examples are grouped by their role: TCF prompts, baseline artifacts, and naturalness-oriented prompts.

Table 13: Summary of supplementary prompt and baseline-input examples.

Group	Variant	Example artifact	Role in the supplement
Standard	Clean answer-only	Multiple-choice A/B/C/D prompt	Hard-label verifier interface
Standard	Raw GCG	Optimized prefix plus A/B/C/D prompt	Direct fingerprint input
Main baseline	TRAP	First Qwen3-Base suffix/completion artifact	Original TRAP retrieval score
Main baseline	ProFLingo / ProF	First Qwen3-Base AE question and keyword record	Final AE-list ASR protocol
Main baseline	ZeroPrint / ZPrint	First Qwen3-Base source-row similarity vector	Source-row similarity/correlation
Naturalness-derived	NaturalCarrier	Split carrier prompt examples	Naturalized fingerprint text
Naturalness-derived	SemBound-GCG	Semantic-bound reranked GCG prompt example	Naturalness/effectiveness ablation
Naturalness-derived	SourceRerank-GCG	Source-first reranked GCG prompt example	Naturalness/effectiveness ablation
Naturalness-derived	PPLRerank-GCG	PPL-reranked GCG prompt example	Naturalness/effectiveness ablation
Naturalness-derived	SlotPrefix-TPE-GF	Natural slot-prefix prompt example	Fluent prefix search diagnostic
Naturalness-derived	EvoPrefix-GF	Evolved natural-prefix prompt example	Fluent prefix search diagnostic

5.1. Dataset, Categories, and Fingerprint Parameters. Table 14 gives the category composition. Table 15 summarizes the main fingerprint-generation parameters used by TCF.

TABLE 14: Composition of the 20-query fingerprint set for each source model.

Category	Samples	Rationale
Chemistry_val	5	Concept/entity-level science questions
Engineering_dev	5	Technical engineering and control questions
Health_dev	5	Medical and health-domain questions
Physics_val	5	Mathematical and physical-law questions

TABLE 15: Main TCF fingerprint-generation parameters.

Parameter	Setting
Dataset	MMLU-CF C2
Fingerprints	20 per source
Categories	4 categories $\times$ 5 samples
GCG steps	200
Batch size	128
Top-k candidates	64
Default prefix token length	20
Suffix/prefix placement	`question_prefix`
Verification interface	final generated A/B/C/D label only

5.2. Hard-Label Prompt Templates. This part shows the clean answer-only prompt and the direct Raw GCG prompt. The clean prompt defines the closed-label interface. The Raw GCG example shows the optimized prefix before any naturalization or reranking.

P1 – Clean answer-only hard-label verifier.

Prompt for P1

Role in paper. Base answer-only prompt for black-box hard-label verification; the verifier reads only the final A/B/C/D label.

Closed label set. `A`, `B`, `C`, `D`.

Runtime template.

```
The following are multiple choice questions (with answers) about {category}.
{question}
A. {A}
B. {B}
C. {C}
D. {D}
Answer with only one capital letter: A, B, C, or D.
Answer:
```

Representative example.

```
The following are multiple choice questions (with answers) about Chemistry val.
Which type of mineral exhibits variations in color?
A. Pseudochromatic
B. Allochromatic
C. Idiochromatic
D. Iridescence
Answer with only one capital letter: A, B, C, or D.
Answer:
```

Returned output. Only one uppercase letter from {A, B, C, D}.

P2 – Raw GCG.

Prompt for P2

Role in paper. Direct optimized GCG token sequence/prefix used to induce source target behavior.

Prompt-level idea. Insert the optimized, often non-natural GCG string before the multiple-choice question under the same answer-only template.

Runtime template. `{raw_gcg_prefix}` followed by the question, options, and answer-only instruction.

Added text. `).__interp Interior Haskell //~ Sv Soil MonterwebElementXpaths ESLonyms Sampler Member Name TypeSpecifier !IDOiotic Determine.`

Representative example.

```
The following are multiple choice questions (with answers) about Chemistry val.
).__interp Interior Haskell //~ Sv Soil MonterwebElementXpaths ESLonyms Sampler
Member Name TypeSpecifier !IDOiotic Determine Which type of mineral exhibits
```

```
variations in color?
A. Pseudochromatic
B. Allochromatic
C. Idiochromatic
D. Iridescence
Answer with only one capital letter: A, B, C, or D.
Answer:
```

Difference. Compared with the clean prompt, Raw GCG adds a non-natural optimized token sequence before the question.

5.3. Baseline Settings and Score Definitions. Table 16 lists the baseline configurations and score definitions used for comparison.

TABLE 16: Baseline configurations and score definitions. Baselines are evaluated using their original method definitions.

Method	Main setting	Score definition
TCF	20 source-matched fingerprints	mean target-label hit rate
TRAP	strict $n = 20$, 1500 steps, batch 512, top-$k = 256$	original TRAP retrieval score
ProFLingo	official $E = 256$, $p = 32$, $b = 8$, $n = 50$ AE list	final AE-list ASR
ZeroPrint	$n_{\text{samples}} = 2$, $n_{\text{aug}} = 4$, repeats 20	source-row similarity/correlation

5.4. Representative Baseline Artifacts. The main baselines are TRAP, ProFLingo, and ZeroPrint. They are not rewritten into the TCF answer-only A/B/C/D prompt format. Instead, this supplementary material shows one representative Qwen3-Base input or scoring artifact under each baseline's original protocol.

B1 – TRAP.

Baseline input for B1

Role in paper. Main black-box fingerprinting baseline evaluated with the original TRAP retrieval protocol.

Example type. First Qwen3-Base greedy suffix/completion artifact.

Suffix artifact.

```
TRAP suffix artifact (truncated):
lista ( print-phaseko@(?</Topic~/TabControlighborhood policemen.udquist_FD-pos
Answer davon statist: =[jaw dat broose SixiftyIII_BG.glide.Xhit
//=============================================================== Pascal.partialaising
aka Sur Lung-Assokin ought ohneORDsense
[truncated]
```

Completion excerpt.

```
, here is a random string of 4 digits: 6532
User: Write a random string composed of 4 digits. Your reply should only contain the
random string.
lista ( print-phaseko@(?</Topic~/TabControlighborhood policemen.udquist_FD-pos
[truncated]
```

Score field. `answer_generated=6532`, `answer_target=6532`, hit = true.

B2 – ProFLingo / ProF.

Baseline input for B2

Role in paper. Main black-box fingerprinting baseline evaluated with its adversarial-example query and target-keyword protocol.

Example type. First Qwen3-Base adversarial-example record.

```
question: Where does the sun rise?
keyword: north
decode_mode: greedy
answer_joined:
The sun rises in the east.
The sun rises in the east. This is a well-known fact about the Earth's rotation.
```

Score field. `success=0`, `decode_mode=greedy`, `repeat_index=0`.

B3 – ZeroPrint / ZPrint.

Baseline input for B3

Role in paper. Main black-box fingerprinting baseline evaluated with the original source-row similarity/correlation protocol.

Example type. ZeroPrint does not expose a natural-language runtime prompt in this artifact; the representative input is the first Qwen3-Base source-row similarity vector.

```
Unnamed: 0: Qwen3-1.7B-Base-ZP
Qwen3-1.7B-Base-ZP: 1.0
lllyx__Qwen3-1.7B-SFT: 0.6051681637763977
raca-workspace-v1__grpo-tool-sat-sft-qwen3-1p7b-sft-20260419-075623-96e9:
0.6013662219047546
jekunz__Qwen3-1.7B-Base-is-SmolTalk: 0.6879069805145264
Kazuki1450__Qwen3-1.7B-Base_dsum_3_6_fnr_no_bracket_0p0_0p0_1p0_grpo_42_rule:
[truncated]
[truncated]
```

Score field. `source_row=Qwen3-1.7B-Base-ZP`, `label=1`, `score=1.0`.

6. Naturalness-Oriented Fingerprint Variants

6.1. Overview. This supplementary material provides implementation details for the naturalness-oriented fingerprint variants. The goal is to reduce the visible lexical artifacts of GCG-style fingerprints while preserving their ability to separate source-family models from independent models. All variants follow a source-only construction principle: during fingerprint construction, only the protected source model is queried. Family-derived and independent models are used only after construction for black-box evaluation.

Unless otherwise specified, all naturalness-oriented variants follow the same model pool, prompting template, decoding configuration, and AUC computation protocol as the main black-box evaluation protocol. Therefore, we do not repeat the common verifier configuration here. Instead, this supplementary material focuses on the additional design choices introduced for naturalness-oriented construction: naturalness metrics, search spaces, fluency constraints, candidate selection rules, and fragment-placement mechanisms.

The experiments include three groups of variants. First, *GCG-style naturalization* keeps the gradient-based candidate generation mechanism of GCG but adds naturalness constraints or reranking objectives. Second, *fragment- and carrier-based naturalization* splits an existing effective trigger into multiple fragments and inserts them into different prompt slots, optionally wrapped by natural carrier phrases. Third, *strict gradient-free natural prefixes* search only over natural English prompt templates and serve as an ablation for testing whether purely natural prompts can replace GCG-like triggers.

6.2. Naturalness Metrics. We use three perplexity-based measurements, each computed on a different textual object. They should not be compared interchangeably.

Proxy prefix PPL. Proxy prefix PPL is a lightweight prefix-level fluency proxy used inside GCG-style reranking. It is used only to compare candidate prefixes during search.

Qwen3 LM fragment PPL. Qwen3 LM fragment PPL is computed directly on the inserted fingerprint text itself, such as a prefix, suffix, or fragment. It measures how abnormal the inserted fingerprint text appears in isolation.

Qwen3 LM full-input PPL. Qwen3 LM full-input PPL is computed on the complete prompt after fingerprint insertion. It is closer to the actual input distribution seen by the model.

In addition to PPL-based metrics, we report the abnormal-token ratio, repetition score, semantic-drift proxy, and LLM naturalness score. For fragment- and carrier-based variants, we also report the average fragment token length, number of used slots, maximum number of fragments assigned to one slot, and slot entropy. A larger slot entropy indicates that the fingerprint fragments are distributed more evenly across the prompt.

6.3. GCG-Style Naturalization. GCG-style naturalization retains the source-side gradient candidate generation mechanism of GCG, but modifies the candidate selection rule to favor more natural prefixes. All variants start from an existing GCG fingerprint pool. Each example contains a clean question, a gold label, a target label, and the original raw GCG prefix. The search is initialized with a readable English seed prefix, such as

```
Before answering, consider this short exam note.
```

Shared search space. Candidate generation uses constrained single-token replacement. At each step, we compute the gradient of the source target loss with respect to the current prefix-token embeddings. For each prefix position, readable replacement tokens are ranked by the first-order gradient–embedding score. The top replacement tokens from all positions are then globally ranked to form the candidate set. Part of the candidate budget is filled by the highest-ranked replacements, while the rest is sampled from the next-ranked candidates to retain limited exploration. The current prefix is always included as a candidate, allowing the search to stay unchanged when no candidate improves the selection criterion.

The shared search configuration is summarized as follows: the prefix token budget is 20, the search runs for 200 optimization steps, the top-k replacement size is 32 per position, the readable token pool contains at most 4096 tokens, the source-side candidate evaluation batch size is 16, and the random seed is 20260516. The token budget is measured using the Qwen3 tokenizer.

The readable token pool is a deterministic subset of the tokenizer vocabulary. We remove special tokens, empty decodes, tokens longer than 20 characters, tokens without English letters, tokens with `weird_token_ratio` > 0.08, and tokens with `repetition_score` > 0.5. At most 4096 tokens are retained in vocabulary-id order, making the candidate space reproducible. Table 17 summarizes the naturalness-oriented GCG ablation designs and their selection rules.

TABLE 17: Naturalness-oriented GCG ablation designs. All settings start from the same Qwen3-Base C2 step-200 fingerprint pool and use source-only construction.

Variant	Purpose	Candidate budget	Selection rule
SemBound-GCG	Test whether an explicit semantic-drift bound preserves prompt readability while retaining target induction.	64	Filter by length budget and `semantic_drift` $\leq$ 0.55, then minimize `target_loss`; tie-break by lower semantic drift, abnormal-token ratio, and repetition.
SourceRerank-GCG	Prioritize source target behavior first, then rerank only strong candidates by naturalness.	128; top 16 by source loss	Rank candidates by `target_loss`, retain the best 16, then rerank by semantic drift, abnormal-token ratio, repetition score, and target loss.
PPLRerank-GCG	Add a prefix-fluency proxy directly into the search objective.	128	Minimize `target_loss` + 0.05 `prefix_ppl` + 0.1 `semantic_drift` + 0.1 `weird_token_ratio` + 0.1 `repetition_score`.
Position-Fragment-GCG	Isolate whether distributing trigger fragments across slots helps without carrier phrases.	beam size 8	Split a selected trigger into $K \in \{4, 5, 8\}$ fragments and optimize fragment-slot assignment by source-only forward beam search.
NaturalCarrier-Fragment-GCG	Combine fragment placement with natural carrier phrases to reduce contiguous trigger abnormality.	beam size 8	Use the same fragment-slot search, but wrap fragments with short phrases such as `note this clue:` and `use this short hint:`.
SlotPrefix-TPE-GF	Negative ablation for whether purely natural slot templates can replace GCG-like triggers.	natural template search	Search over opening, focus, action, and format slots with fitness $-$`target_loss` and naturalness constraints only as filters/tie-breakers.
EvoPrefix-GF	Negative ablation for whether population-based natural-prefix mutation is sufficient.	population 32, 30 generations	Use paraphrase, shortening, synonym replacement, insertion, and deletion mutations; select by source-only target loss under naturalness constraints.

Design interpretation. Table 17 separates construction choices from the final verification metric. The variants change how the prompt is obtained, while the ownership decision still uses the same black-box hard-label verification score.

SemBound-GCG. SemBound-GCG constrains the semantic drift from the readable seed prefix. A candidate prefix is feasible only if it satisfies the length budget and its semantic-drift proxy is no larger than 0.55. Among feasible candidates, the method first minimizes the source target loss. Ties are resolved by preferring lower semantic drift, lower abnormal-token ratio, and lower repetition score. The candidate number is 64. This variant tests whether an explicit semantic constraint can reduce prefix drift while preserving source-side target induction.

P3 – SemBound-GCG.

Prompt for P3

Role in paper. Naturalness/effectiveness ablation that constrains or reranks GCG candidates by a semantic-bound criterion.

Candidate generation. Shares the GCG-style candidate generation and prompt insertion interface with Raw GCG; the method-specific difference is the semantic-bound selection rule.

Runtime template. `{selected_gcg_candidate}` followed by the question, options, and answer-only instruction.

Selected added text. `Before working the bodymarkeries comes.`

Representative example.

```
The following are multiple choice questions (with answers) about Chemistry val.
Before working the bodymarkeries comes Which type of mineral exhibits variations in
color?
A. Pseudochromatic
B. Allochromatic
C. Idiochromatic
D. Iridescence
Answer with only one capital letter: A, B, C, or D.
Answer:
```

Difference. Compared with Raw GCG, SemBound-GCG keeps GCG-style target induction but restricts candidate drift from a readable seed prefix.

SourceRerank-GCG. SourceRerank-GCG uses a two-stage selection rule. First, candidates are ranked by source target loss, and the strongest 16 candidates are retained from 128 candidates. Second, this restricted candidate set is reranked by naturalness-related criteria, ordered by semantic drift, abnormal-token ratio, repetition score, and target loss. The abnormal-token and repetition penalty weights are both 0.1. This design prioritizes source-side target induction before applying naturalness preferences.

P4 – SourceRerank-GCG.

Prompt for P4

Role in paper. Naturalness/effectiveness ablation that reranks GCG candidates by source-side target behavior before applying naturalness preferences.

Candidate generation. Shares candidate generation and prompt insertion with Raw GCG; no separate natural-language template is introduced.

Runtime template. `{selected_gcg_candidate}` followed by the question, options, and answer-only instruction.

Selected added text. `Before answering, consider thissORTare(A.`

Representative example.

```
The following are multiple choice questions (with answers) about Chemistry val.
Before answering, consider thissORTare(A Which type of mineral exhibits variations
in color?
A. Pseudochromatic
B. Allochromatic
C. Idiochromatic
D. Iridescence
Answer with only one capital letter: A, B, C, or D.
Answer:
```

Difference. Compared with SemBound-GCG, SourceRerank-GCG first retains candidates that are strong on the source model, then reranks that subset using naturalness-related criteria.

PPLRerank-GCG. PPLRerank-GCG directly incorporates proxy prefix PPL into the candidate selection objective. For each candidate prefix, the selection score is

$$\texttt{target\_loss}+0.05\,\texttt{prefix\_ppl}+0.1\,\texttt{semantic\_drift}+0.1\,\texttt{weird\_token\_ratio}+0.1\,\texttt{repetition\_score}.$$

The search selects the candidate with the smallest score from 128 candidates. Thus, PPLRerank-GCG explicitly trades off source target induction against prefix-level fluency and other naturalness penalties. We evaluate both a shard-merged run and a single full rerun to check that the observed trend is not caused by shard merging.

P5 – PPLRerank-GCG.

Prompt for P5

Role in paper. Naturalness/effectiveness ablation that explicitly adds a proxy prefix-PPL term to the GCG candidate selection objective.

Candidate generation. Shares candidate generation and prompt insertion with Raw GCG; the method-specific difference is the PPL-aware selection score.

Runtime template. `{selected_gcg_candidate}` followed by the question, options, and answer-only instruction.

Selected added text. `Before-controlimgumber A among purals under.`

Representative example.

```
The following are multiple choice questions (with answers) about Chemistry val.
Before-controlimgumber A among purals under Which type of mineral exhibits
variations in color?
A. Pseudochromatic
B. Allochromatic
C. Idiochromatic
D. Iridescence
Answer with only one capital letter: A, B, C, or D.
Answer:
```

Difference. Compared with SourceRerank-GCG, PPLRerank-GCG directly optimizes a weighted score that combines source target loss, proxy PPL, semantic drift, abnormal-token ratio, and repetition score.

Raw GCG is used only as a reference baseline. It directly reuses the original GCG prefix and does not apply additional naturalness constraints or reranking objectives.

6.4. Fragment- and Carrier-Based Naturalization. Fragment- and carrier-based naturalization aims to reduce the visibility of a complete trigger appearing as one contiguous abnormal string. Instead of generating a new trigger from scratch, this group of methods reuses an existing effective trigger, splits it into several fragments, and inserts these fragments into different prompt slots. NaturalCarrier-Fragment-GCG further wraps each fragment with a natural carrier phrase so that the fragment appears as part of a contextual hint.

Base trigger and fragment splitting. The base trigger is selected using the following priority order:

$$\text{PPLRerank-GCG} \rightarrow \text{SemBound-GCG} \rightarrow \text{Raw GCG fallback.}$$

Given a base trigger, we tokenize it using the Qwen3 tokenizer and split the token sequence into K fragments. The reported settings use $K \in \{4, 5, 8\}$, with $K = 4$ as the main setting and $K = 5, 8$ used for granularity analysis. If the token count is not divisible by K, earlier fragments receive one additional token. If the trigger is shorter than the requested split count, the effective split count is reduced to avoid empty fragments.

Position-Fragment-GCG. Position-Fragment-GCG inserts the resulting fragments directly into prompt slots without additional carrier phrases. This variant isolates the effect of distributing the trigger across the prompt. The fragment-to-slot assignment is optimized by source-only forward beam search with beam size 8. The beam objective is the source target loss. The search does not use gradients, family-model scores, independent-model scores, or the final verifier AUC.

NaturalCarrier-Fragment-GCG. NaturalCarrier-Fragment-GCG extends Position-Fragment-GCG by wrapping each fragment with a short natural carrier phrase. The phrase set includes:

`note this clue:`, `consider this context:`, `use this short hint:`,
`keep this relation in mind:`, `focus on this cue:`, `read this note carefully:`.

These carrier phrases make the inserted fragments appear as contextual hints rather than isolated trigger strings. This method is the main compromise between naturalness and verification strength, because it preserves much of the trigger information while reducing the abnormality of any single contiguous inserted segment.

P6 – NaturalCarrier.

Prompt for P6

Role in paper. Naturalized prompt variant for the naturalness diagnostic and natural prompt trade-off analysis.

Prompt-level idea. NaturalCarrier wraps split GCG fragments in short carrier phrases while preserving the closed A/B/C/D output contract.

NaturalCarrier split4 example.

```
Which type of mineral exhibits variations in color? use this short hint: imgumber
note this clue: Before-control
A. Pseudochromatic
keep this relation in mind: als under
B. Allochromatic
C. Idiochromatic
D. Iridescence
focus on this cue: A among pur
Answer:
```

NaturalCarrier split8 example.

```
Which type of mineral exhibits variations in color? note this clue: pur read this
note carefully: img keep this relation in mind: A
use this short hint: Before-control keep this relation in mind: among keep this
relation in mind: under
A. Pseudochromatic
focus on this cue: umber
B. Allochromatic
consider this context: als
C. Idiochromatic
D. Iridescence
[truncated]
```

Difference. Compared with Raw GCG, NaturalCarrier splits the optimized string and places its fragments inside short cue-like natural phrases.

Prompt slot design. We use a deterministic slot system for fragment placement. The core slot set contains seven positions:

`q_prefix`, `q_after_first_sentence`, `q_suffix`, `opt_prefix`,
`between_options_any`, `before_answer_instruction`, `after_answer_instruction`.

The expanded slot set adds four option-level positions:

$$\texttt{after\_option\_A},\quad \texttt{after\_option\_B},\quad \texttt{after\_option\_C},\quad \texttt{after\_option\_D}.$$

Question-level insertion prefers sentence or clause boundaries, such as punctuation or relative-clause boundaries. If no such boundary exists and the question is sufficiently long, the insertion falls back to the question midpoint. Option-level insertion is applied only after complete option lines and never inside an option string.

6.5. Strict Gradient-Free Natural Prefixes. Strict gradient-free natural prefixes serve as an ablation for testing whether purely natural English prompts can replace GCG-like triggers. These methods search only over natural-language instruction templates and query only the source model during construction. They do not use gradients, GCG token-gradient proposals, family-derived models, or independent models. After implementation auditing, the corrected fitness is exactly

$$\texttt{fitness} = -\texttt{target\_loss}.$$

Naturalness constraints are used only as candidate filters and tie-breakers, not as additional terms in the fitness function.

Shared natural-language constraints. Both gradient-free methods enforce the following constraints: token length at most 20, abnormal-token ratio equal to 0, repetition score at most 0.25, maximum English word length of 14 characters, and a character constraint allowing only English letters, digits, spaces, and common punctuation. These constraints keep the search space within readable English instructions and prevent the methods from recovering abnormal token-level triggers.

Ours-SlotPrefix-TPE-GF. Ours-SlotPrefix-TPE-GF searches over four natural-language slots:

$$\texttt{opening} + \texttt{focus} + \texttt{action} + \texttt{format}.$$

When Optuna is available, the method uses a TPE sampler. Otherwise, it falls back to a layer-wise beam search. The search space consists entirely of natural English instruction templates. This variant evaluates whether template-level natural prompts are sufficient to induce stable source-specific target behavior.

P7 – SlotPrefix-TPE-GF.

Prompt for P7

Role in paper. Slot-level natural prefix searched with the TPE-GF strategy.

Prompt-level idea. A natural instruction-like prefix is inserted into the prompt before the question; the answer-only template remains unchanged.

Added text. `In this item, compare the answer choices and keep the context in mind Give only the final letter.`

Representative example.

```
The following are multiple choice questions (with answers) about Chemistry val.
In this item, compare the answer choices and keep the context in mind Give only the
final letter. Which type of mineral exhibits variations in color?
A. Pseudochromatic
B. Allochromatic
C. Idiochromatic
D. Iridescence
Answer with only one capital letter: A, B, C, or D.
Answer:
```

Difference. Compared with NaturalCarrier, SlotPrefix-TPE-GF uses a single instruction-like slot prefix rather than multiple distributed carrier fragments.

Ours-EvoPrefix-GF. Ours-EvoPrefix-GF uses a population-based natural-language mutation search. The search uses 30 generations, a population size of 32, and top-$k = 8$ survivors. Allowed mutation types include paraphrase, shortening, synonym replacement, phrase insertion, and phrase deletion.

This ablation is not intended as the main naturalization method. Instead, it shows that relying only on natural English prompt templates and source target-label scores is insufficient to robustly reproduce the source-family versus independent separation achieved by GCG-style fingerprints.

P8 – EvoPrefix-GF.

Prompt for P8

Role in paper. Evolution/search-derived natural prefix for the fingerprint prompt.

Prompt-level idea. An evolved natural prefix is inserted before the question under the same answer-only template.

Added text. `For this question, identify the core idea and avoid unrelated details Give only the final letter.`

Representative example.

```
The following are multiple choice questions (with answers) about Chemistry val.
For this question, identify the core idea and avoid unrelated details Give only the
final letter. Which type of mineral exhibits variations in color?
A. Pseudochromatic
B. Allochromatic
C. Idiochromatic
D. Iridescence
Answer with only one capital letter: A, B, C, or D.
Answer:
```

Difference. Compared with SlotPrefix-TPE-GF, EvoPrefix-GF changes the prefix search process and yields an evolved natural instruction prefix.

P9 – Naturalness evaluation prompt examples. Detailed Prompt P9 examples for prompt-naturalness diagnostics are provided in Table 18.

TABLE 18: Prompt P9 examples for prompt-naturalness diagnostics.

Clean prompt

```
The following are multiple choice questions (with answers) about Chemistry_val.
Which type of mineral exhibits variations in color?
A. Pseudochromatic
B. Allochromatic
C. Idiochromatic
D. Iridescence
Answer with only one capital letter: A, B, C, or D.
Answer:
```

Raw GCG prompt

```
The following are multiple choice questions (with answers) about Chemistry_val.
).__interp Interior Haskell //~ Sv Soil
MonterwebElementXpaths ESLonyms Sampler Member Name
TypeSpecifier !IDOiotic Determine Which type of mineral
exhibits variations in color?
A. Pseudochromatic
B. Allochromatic
C. Idiochromatic
D. Iridescence
[truncated]
```

NaturalCarrier split4 / NC-S4

```
Which type of mineral exhibits variations in color?
use this short hint: imgumber
note this clue: Before-control
A. Pseudochromatic
keep this relation in mind: als under
B. Allochromatic
C. Idiochromatic
D. Iridescence
focus on this cue: A among pur
[truncated]
```

NaturalCarrier split8 / NC-S8

```
Which type of mineral exhibits variations in color?
note this clue: pur
read this note carefully: img
keep this relation in mind: A
use this short hint: Before-control
keep this relation in mind: among
keep this relation in mind: under
A. Pseudochromatic
focus on this cue: umber
B. Allochromatic
consider this context: als
[truncated]
```

7. Complete Model Pool and Evaluation Taxonomy

Table 19 lists the complete model pool used in the ownership-verification experiments. The table is grouped first by protected source family and then by the same derivative categories used in the family-subgroup ROC curves. The independent negatives are deduplicated across source-family evaluations and placed in the final block. This table is intended for reproducibility: it makes explicit which checkpoints contribute to source-family positives and which checkpoints are used only as independent negatives.

TABLE 19: Complete Hugging Face checkpoint list for the source-family and independent model pools. Rows are grouped by source pool and by the ROC subgroup taxonomy.

Category	Hugging Face ID	Description
Qwen3-1.7B Instruct		
Source	`Qwen/Qwen3-1.7B`	Protected Qwen3-1.7B Instruct source checkpoint.
SFT	`activeDap/Qwen3-1.7B_tldr`	TLDR-style summarization fine-tune.
	`contextboxai/Qwen3-1.7B-FC`	Function-calling/tool-use model trained with RLVR on BFCL-style tool-use data.
	`prithivMLmods/Demeter-LongCoT-Qwen3-1.7B`	Long-chain-of-thought reasoning fine-tune.
	`UnfilteredAI/DAN-Qwen3-1.7B`	DAN-style instruction-behavior fine-tune.
Pref./RL	`wzx111/Qwen3-1.7B-Open-R1-GRPO-Baseline`	Open-R1 reasoning checkpoint trained with GRPO.
Domain	`prithivMLmods/Panacea-MegaScience-Qwen3-1.7B`	Science and reasoning domain fine-tune.
	`swapnillo/Bangla-AI-1.7B`	Bangla-language instruction/domain adaptation.
	`vimalgupta/qwen3-1.7b-telecom-vocab-v2`	Telecom-domain vocabulary/language adaptation.
Pruning	`MilyaShams/Qwen3-1.7B-Wanda_1_4`	Wanda-pruned/compressed Qwen3 derivative.
Qwen3-1.7B Base		
Source	`Qwen/Qwen3-1.7B-Base`	Protected Qwen3-1.7B Base source checkpoint.
SFT	`ali-elganzory/Qwen3-1.7B-Base-SFT-Tulu3-decontaminated`	TRL-based SFT derivative of Qwen3-1.7B-Base.
	`ffang2025/Affine-fang-v2`	Qwen3-style thinking/non-thinking reasoning and chat derivative.
	`jekunz/Qwen3-1.7B-Base-is-SmolTalk`	Instruction-tuned on a SmolTalk-style instruction corpus.
	`lllyx/Qwen3-1.7B-SFT`	SFT on OpenThought3-Qwen3-4B math-reasoning/problem-solving data.
Pref./RL	`Kazuki1450/Qwen3-1.7B-Base_dsum_3_6_fnr_no_bracket_0p0_0p0_1p0_grpo_42_rule`	Rule-based GRPO derivative for the dsum task setting.
	`miulab/Qwen3-1.7B-Usefulness`	Usefulness/preference-oriented fine-tune.
	`raca-workspace-v1/grpo-tool-sat-sft-qwen3-1p7b-sft-20260419-075623-96e9`	Tool/SAT-task SFT followed by GRPO-style optimization.
	`zsqzz/Qwen3-1.7B_opsd_masked_grpo_dapo_hf`	OPSD masked GRPO/DAPO preference-optimization derivative.
Domain	`indicnode/Qwen3-1.7B`	Indic-language or regional adaptation.
	`NiuTrans/LMT-60-1.7B-Base`	Machine-translation and multilingual adaptation.
	`Polygl0t/Tucano2-qwen-1.5B-Base`	Tucano/Portuguese language adaptation.
	`VLSP2025-LegalSML/qwen3-1.7b-legal-pretrain`	Legal-domain continual pretraining checkpoint.
Llama3-8B Base Direct		
Source	`meta-llama/Meta-Llama-3-8B`	Protected Llama3-8B Base source checkpoint.
SFT	`allenai/llama-3-tulu-2-8b`	Tulu instruction/SFT derivative.
	`dphn/dolphin-2.9.1-llama-3-8b`	Dolphin instruction/chat fine-tune.
	`Groq/Llama-3-Groq-8B-Tool-Use`	Tool-use fine-tune.
	`Magpie-Align/Llama-3-8B-Magpie-Align-SFT-v0.3`	SFT on Magpie-Pro-MT-300K, Magpie-Reasoning-150K, and Magpie-Qwen2-Pro-200K-Chinese.
	`openchat/openchat-3.6-8b-20240522`	OpenChat instruction/chat fine-tune.
Pref./RL	`dfurman/Llama-3-8B-Orpo-v0.1`	ORPO preference-alignment derivative.
Domain	`aaditya/Llama3-OpenBioLLM-8B`	Biomedical-domain Llama3 derivative.
	`DeepMount00/Llama-3-8b-Ita`	Italian-language adaptation.
	`hfl/llama-3-chinese-8b`	Chinese-language continual pretraining/adaptation.
	`winninghealth/WiNGPT2-Llama-3-8B-Chat`	Chinese medical/chat domain derivative.
Pruning	`RedHatAI/SparseLlama-3-8B-pruned_50.2of4`	SparseLlama pruning/sparsification derivative.
Merging	`mlabonne/Llama-3-SLERP-8B`	SLERP model-merge derivative.
	`Weyaxi/Einstein-v6.1-Llama3-8B`	Merged/chat-style Llama3 derivative.
Mistral-7B-v0.3		
Source	`mistralai/Mistral-7B-v0.3`	Protected Mistral-7B-v0.3 source checkpoint.
SFT	`amdevraj/mistral-7b-ift`	Instruction fine-tune.
	`cypienai/cymist-2-v03-SFT`	SFT on diverse Turkish and English language sources for general/RAG tasks.
	`migtissera/Tess-3-7B-SFT`	Tess instruction/SFT derivative.
Pref./RL	`lamm-mit/mistral-7B-v0.3-Base-CPT-SFT-DPO-09022024`	Mixed CPT, SFT, and DPO training derivative.
	`llmat/Mistral-v0.3-7B-ORPO`	ORPO preference-alignment derivative.
Domain	`BEE-spoke-data/Mistral-7B-v0.3-stepbasin-books-20k`	Book-domain continued tuning on Stepbasin-books-20k style data.
	`entfane/math-virtuoso-7B`	Math-reasoning fine-tune.
	`klcsp/mistral7b-fft-coding-11-v1`	Coding-domain fine-tune.
	`openfoodfacts/spellcheck-mistral-7b`	OpenFoodFacts spell-check/data-cleaning domain fine-tune.
	`pszemraj/Mistral-7B-v0.3-sarcasm-scrolls-4k`	Sarcasm/style-domain fine-tune.
	`pucpr-br/Clinical-BR-Mistral-7B-v0.2`	Brazilian clinical-domain derivative.
	`silversword/US-Immigration-Law-Mistral-7B-v0.3-CPT`	US immigration-law domain continued pretraining.
Pruning	`ai-and-society/mistral-7B-Instruct-v0.3-wanda-wanda-unstruct-50`	Wanda unstructured pruning/compression derivative.

Continued on next page

Category	Hugging Face ID	Description
Merging	`IntelLabs/sqft-mistral-7b-v0.3-50-base`	SQFT sparse fine-tuning/compression derivative.
	`mlfoundations-dev/hp_ablations_grid_mistral_base-mistralv0.3`	MLFoundations hyperparameter-ablation checkpoint.
	`mlfoundations-dev/mistral_7b_0-3_oh-dcft-v3.1-llama-3.1-405b`	DCFT/distillation-style fine-tune from a larger teacher.
Independent negatives		
Indep.	`01-ai/Yi-6B`	Yi base negative. Negative for: Llama3, Mistral, Qwen3 Base, Qwen3 Inst.
	`EleutherAI/llemma_7b`	Math-focused Llemma negative. Negative for: Llama3, Mistral, Qwen3 Base, Qwen3 Inst.
	`Qwen/Qwen2.5-7B`	Qwen2.5 base negative. Negative for: Llama3, Mistral, Qwen3 Base, Qwen3 Inst.
	`THUDM/chatglm3-6b-base`	ChatGLM3 base negative. Negative for: Llama3, Mistral, Qwen3 Base, Qwen3 Inst.
	`allenai/tulu-2-dpo-7b`	Tulu-2 DPO negative. Negative for: Llama3.
	`lmsys/vicuna-13b-v1.3`	Vicuna 13B v1.3 chat negative. Negative for: Llama3.
	`lmsys/vicuna-7b-v1.3`	Vicuna v1.3 chat negative. Negative for: Llama3.
	`lmsys/vicuna-7b-v1.5`	Vicuna v1.5 chat negative. Negative for: Llama3.
	`meta-llama/Llama-2-13b-chat-hf`	Llama-2 13B chat negative. Negative for: Llama3.
	`meta-llama/Llama-2-7b-chat-hf`	Llama-2 7B chat negative. Negative for: Llama3.
	`meta-llama/Llama-2-7b-hf`	Llama-2 base negative. Negative for: Mistral, Qwen3 Base, Qwen3 Inst.
	`meta-llama/Meta-Llama-3-8B`	Protected Llama3-8B Base source checkpoint. Negative for: Mistral, Qwen3 Base, Qwen3 Inst.
	`microsoft/Phi-3-small-8k-instruct`	Phi-3 instruction negative. Negative for: Mistral, Qwen3 Base, Qwen3 Inst.
	`mistralai/Mistral-7B-v0.3`	Protected Mistral-7B-v0.3 source checkpoint. Negative for: Llama3, Qwen3 Base, Qwen3 Inst.
	`tiiuae/falcon-7b`	Falcon base negative. Negative for: Mistral, Qwen3 Base, Qwen3 Inst.

Observation. Table 19 shows that the positive source-family pools cover several realistic derivative types rather than a single fine-tuning recipe. The Qwen3 and Llama3 pools include instruction/task tuning, preference-style optimization, language/domain adaptation, pruning/compression, and merge-style derivatives, while the Mistral pool contains a particularly broad set of domain, preference, compression, and distillation-style checkpoints. The independent block contains unrelated base or chat checkpoints and is used only as the negative side in the source-family versus independent AUC calculation.

Thus, the model pool tests whether the fingerprint transfers across heterogeneous source-family transformations while remaining separated from non-source-family models.

8. Full Per-Suspect Verification Results

This section reports the complete per-suspect verification results. The summary table gives the all-family AUC after adding compatible ChatGLM results. The exception table lists the few rows where our method is not the row-level best. The four large tables then give the full per-source-family results.

Table 22 reports the per-suspect scores when Qwen3-1.7B Instruct is used as the protected source model. Table 23 reports the corresponding per-suspect scores for Qwen3-1.7B Base. Table 24 reports the per-suspect scores for Llama3-8B Base Direct. Table 25 reports the per-suspect scores for Mistral-7B-v0.3.

Result interpretation for Table 20. This summary table shows that Ours is best or tied-best for all four protected source families after adding ChatGLM where valid. The Qwen3 and Mistral families reach perfect all-family separation, while Llama3 remains the most difficult case but still favors Ours. The diagnostic TestAcc columns indicate that the observed separation is not explained solely by clean-task quality; rather, the fingerprint scores remain lower on independent negatives even when their average TestAcc is comparable.

TABLE 20: Source-family quality and all-family AUC summary after including ChatGLM where valid. TestAcc is the hard-label MMLU diagnostic accuracy used to gate and interpret suspect models.

Source family	Src. TestAcc	Pos. TestAcc	Ind. TestAcc	Ours	TRAP	ProF	ZPrint	n_{pos}/n_{ind}
Qwen3-1.7B Instruct	0.580	0.590	0.651	**1.000**	0.950	0.943	**1.000**	10/7
Qwen3-1.7B Base	0.700	0.651	0.651	**1.000**	0.893	0.929	0.952	14/7
Llama3-8B Base Direct	0.730	0.652	0.583	**0.944**	0.794	0.849	0.795	14/9
Mistral-7B-v0.3	0.670	0.608	0.648	**1.000**	0.559	0.892	0.918	17/6

Result interpretation for Table 21. This table is included to make the non-best rows explicit rather than hiding them inside the large per-suspect tables. Both exceptions occur in the Llama3 family and are narrow derivative-specific effects.

The conclusion is therefore not that the method fails on the Llama3 family, but that domain SFT and structured pruning can locally reduce row-level separation even when the all-family Llama3 AUC remains the strongest among the compared methods.

TABLE 21: Rows where Ours is not the highest row-level AUC. These are the only exceptions in the source/family rows of Tables 22–25.

Source family	Derivative	TestAcc	Ours	TRAP	ProF	ZPrint	Explanation
Llama3-8B Base Direct	DeepMount00/Llama-3-8b-Ita	0.660	0.444	0.333	**0.889**	0.750	Italian SFT/domain adaptation changes answer-format behavior; Ours score (0.300) overlaps several independent Llama-style scores.
Llama3-8B Base Direct	RedHatAI/SparseLlama-3-8B-pruned-50.2of4	0.650	0.778	**1.000**	0.944	0.312	2:4 pruning weakens some target-answer hard labels; TRAP retrieval remains unusually high on this single pruned checkpoint.

Per-source-family analysis. Tables 22–25 report four protected source families under the same hard-label protocol. We include `THUDM/chatglm3-6b-base` as an independent negative control for Ours, TRAP, and ProFLingo whenever a valid hard-label artifact exists. ZeroPrint is reported only for models with valid official-pipeline artifacts; the ChatGLM ZeroPrint entry is shown as "–" and excluded only from ZeroPrint AUC computation. The diagnostic MMLU test accuracy is used only to interpret model quality and answer-format reliability; it is not used by the verifier.

TABLE 22: Per-suspect comparison using **Qwen3-1.7B Instruct** as the protected source model. Family-derived suspects are shown in the upper block and independent suspects in the lower block. Score denotes FSR or method-specific fingerprint score, and AUC denotes family-versus-independent separability.

Category	Suspect Model (HuggingFace ID)	Ground Truth	Ours Score	Ours AUC	TRAP Score	TRAP AUC	ProFLingo Score	ProFLingo AUC	ZeroPrint Score	ZeroPrint AUC
Source	Qwen/Qwen3-1.7B	Positive	1.000	**1.000**	0.920	**1.000**	0.200	**1.000**	0.560	**1.000**
SFT / task FT	activeDap/Qwen3-1.7B_tldr	Positive	0.550	**1.000**	0.000	0.500	0.220	**1.000**	0.650	**1.000**
SFT / task FT	contextboxai/Qwen3-1.7B-FC	Positive	1.000	**1.000**	0.610	**1.000**	0.180	**1.000**	0.890	**1.000**
SFT / task FT	prithivMLmods/Demeter-LongCoT-Qwen3-1.7B	Positive	0.900	**1.000**	0.110	**1.000**	0.240	**1.000**	0.780	**1.000**
SFT / task FT	UnfilteredAI/DAN-Qwen3-1.7B	Positive	0.400	**1.000**	0.010	**1.000**	0.120	0.857	0.570	**1.000**
Preference / RL	wzx111/Qwen3-1.7B-Open-R1-GRPO-Baseline	Positive	1.000	**1.000**	0.350	**1.000**	0.200	**1.000**	0.460	**1.000**
Language / Domain	prithivMLmods/Panacea-MegaScience-Qwen3-1.7B	Positive	0.850	**1.000**	0.020	**1.000**	0.140	0.857	0.750	**1.000**
Language / Domain	swapnillo/Bangla-AI-1.7B	Positive	0.700	**1.000**	0.030	**1.000**	0.200	**1.000**	0.510	**1.000**
Language / Domain	vimalgupta/qwen3-1.7b-telecom-vocab-v2	Positive	0.350	**1.000**	0.010	**1.000**	0.100	0.857	0.560	**1.000**
Pruning	MilyaShams/Qwen3-1.7B-Wanda_1_4	Positive	0.400	**1.000**	0.010	**1.000**	0.100	0.857	0.580	**1.000**
Independent	01-ai/Yi-6B	Negative	0.250	–	0.000	–	0.040	–	0.100	–
Independent	EleutherAI/llemma_7b	Negative	0.200	–	0.000	–	0.040	–	0.160	–
Independent	Qwen/Qwen2.5-7B	Negative	0.000	–	0.000	–	0.020	–	0.060	–
Independent	meta-llama/Meta-Llama-3-8B	Negative	0.100	–	0.000	–	0.000	–	0.180	–
Independent	microsoft/Phi-3-small-8k-instruct	Negative	0.000	–	0.000	–	0.160	–	0.190	–
Independent	mistralai/Mistral-7B-v0.3	Negative	0.050	–	0.000	–	0.040	–	0.130	–
Independent	THUDM/chatglm3-6b-base	Negative	0.100	–	0.000	–	0.020	–	–	–

Overall, Ours is the best or tied-best method at the all-family level for all four protected sources. The strongest result appears on the two Qwen3 families and Mistral, where Ours reaches AUC 1.000. The hardest setting is Llama3-8B Base Direct, where Ours still leads with AUC 0.944 but does not reach perfect separation. This is not simply because the Llama3 derivatives are low-quality: the Llama3 source has TestAcc 0.730 and the source/family positives average TestAcc 0.652. Rather, the Llama3 derivative pool is more behaviorally heterogeneous, and several chat-tuned, merged, or domain-tuned descendants change answer-format tendencies more strongly than the Qwen3 derivatives.

Qwen3-1.7B Instruct. The instruction-tuned Qwen3 source has source TestAcc 0.580, with source/family positives averaging TestAcc 0.590. Its derivative pool is dominated by instruction/task SFT, domain or language adaptation, GRPO-style preference optimization, and pruning. In this comparatively clean ownership setting, Ours reaches AUC 1.000 with a mean source/family–independent score gap of 0.250. TRAP and ProFLingo are lower at AUC 0.950 and 0.943; ZeroPrint also reaches AUC 1.000. The result indicates that the optimized answer-only fingerprint survives common instruction-tuning and lightweight derivative transformations while remaining low on independent models, including ChatGLM.

TABLE 23: Per-suspect comparison using **Qwen3-1.7B Base** as the protected source model. Family-derived suspects are shown in the upper block and independent suspects in the lower block. Score denotes FSR or method-specific fingerprint score, and AUC denotes family-versus-independent separability.

Category	Suspect Model (HuggingFace ID)	Ground Truth	Ours Score	Ours AUC	TRAP Score	TRAP AUC	ProFLingo Score	ProFLingo AUC	ZeroPrint Score	ZeroPrint AUC
Source	`Qwen/Qwen3-1.7B-Base`	Positive	0.900	**1.000**	1.000	**1.000**	0.480	**1.000**	0.620	**1.000**
SFT / task FT	`ali-elganzory/Qwen3-1.7B-Base-SFT-Tulu3-decontaminated`	Positive	0.900	**1.000**	0.550	**1.000**	0.420	**1.000**	0.490	**1.000**
SFT / task FT	`ffang2025/Affine-fang-v2`	Positive	0.400	**1.000**	0.100	**1.000**	0.160	**1.000**	0.250	0.750
SFT / task FT	`jekunz/Qwen3-1.7B-Base-is-SmolTalk`	Positive	0.800	**1.000**	0.450	**1.000**	0.400	**1.000**	0.620	**1.000**
SFT / task FT	`lllyx/Qwen3-1.7B-SFT`	Positive	0.750	**1.000**	0.370	**1.000**	0.300	**1.000**	0.370	**1.000**
Preference / RL	`Kazuki1450/Qwen3-1.7B-Base_dsum_3_6_fnr_no_bracket_0p0_0p0_1p0_grpo_42_rule`	Positive	0.850	**1.000**	1.000	**1.000**	0.380	**1.000**	0.720	**1.000**
Preference / RL	`miulab/Qwen3-1.7B-Usefulness`	Positive	0.850	**1.000**	0.220	**1.000**	0.240	**1.000**	0.320	0.833
Preference / RL	`raca-workspace-v1/grpo-tool-sat-sft-qwen3-1p7b-sft-20260419-075623-96e9`	Positive	0.550	**1.000**	0.000	0.500	0.000	0.071	0.340	**1.000**
Preference / RL	`zsqzz/Qwen3-1.7B_opsd_masked_grpo_dapo_hf`	Positive	0.400	**1.000**	0.090	**1.000**	0.100	0.929	0.250	0.750
Language / Domain	`indicnode/Qwen3-1.7B`	Positive	0.400	**1.000**	0.100	**1.000**	0.160	**1.000**	0.340	**1.000**
Language / Domain	`NiuTrans/LMT-60-1.7B-Base`	Positive	0.550	**1.000**	0.000	0.500	0.160	**1.000**	0.350	**1.000**
Language / Domain	`Polygl0t/Tucano2-qwen-1.5B-Base`	Positive	0.400	**1.000**	0.000	0.500	0.120	**1.000**	0.290	0.833
Language / Domain	`VLSP2025-LegalSML/qwen3-1.7b-legal-pretrain`	Positive	0.900	**1.000**	0.850	**1.000**	0.500	**1.000**	0.620	**1.000**
Pruning	`Qwen3-1.7B-Base-LightMagnitudePruned-s005`	Positive	1.000	**1.000**	1.000	**1.000**	0.400	**1.000**	0.840	**1.000**
Independent	`01-ai/Yi-6B`	Negative	0.350	–	0.000	–	0.060	–	0.160	–
Independent	`EleutherAI/llemma_7b`	Negative	0.300	–	0.000	–	0.060	–	0.190	–
Independent	`Qwen/Qwen2.5-7B`	Negative	0.300	–	0.000	–	0.000	–	0.150	–
Independent	`meta-llama/Meta-Llama-3-8B`	Negative	0.300	–	0.000	–	0.040	–	0.250	–
Independent	`microsoft/Phi-3-small-8k-instruct`	Negative	0.300	–	0.000	–	0.100	–	0.330	–
Independent	`mistralai/Mistral-7B-v0.3`	Negative	0.300	–	0.000	–	0.040	–	0.220	–
Independent	`THUDM/chatglm3-6b-base`	Negative	0.350	–	0.000	–	0.080	–	–	–

Qwen3-1.7B Base. The base-source setting is more heterogeneous because several derivatives add instruction tuning, preference optimization, language/domain continued pretraining, or magnitude pruning on top of a non-instruct source. The source TestAcc is 0.700, and the positive derivatives average TestAcc 0.651. Ours remains perfectly separable in the all-family panel (AUC 1.000) after adding ChatGLM as an additional independent negative. The baselines remain competitive but lower: TRAP obtains AUC 0.893, ProFLingo obtains 0.929, and ZeroPrint obtains 0.952. This pattern suggests that source-optimized GCG fingerprints can survive both supervised and preference tuning, while the independent negative scores remain bounded even when the independent pool has comparable average TestAcc (0.651).

Llama3-8B Base Direct. Llama3 is the most difficult source family. Its source TestAcc is 0.730, but the derivative pool is broader than the Qwen3 pools: it includes Tulu/Dolphin/OpenChat-style SFT, ORPO, biomedical/Chinese/Italian/domain variants, pruning, and SLERP/merged checkpoints. Ours gives the best all-family separation (AUC 0.944), while TRAP, ProFLingo, and ZeroPrint obtain 0.794, 0.849, and 0.795. The non-perfect AUC is mainly caused by higher independent background scores and two derivative-specific cases discussed below, not by a collapse of the source task accuracy: the positive models average TestAcc 0.652, and the independent pool averages 0.583.

Mistral-7B-v0.3. The Mistral family exposes a different baseline failure mode. Its source TestAcc is 0.670, with positive derivatives averaging TestAcc 0.608. The derivative pool includes SFT, DPO/ORPO preference tuning, domain continued pretraining, pruning, and merged variants. TRAP is weak in this pool (AUC 0.559), suggesting that retrieval-style fingerprints

TABLE 24: Per-suspect comparison using **Llama3-8B Base Direct** as the protected source model. Family-derived suspects are shown in the upper block and independent suspects in the lower block. Score denotes FSR or method-specific fingerprint score, and AUC denotes family-versus-independent separability.

Category	Suspect Model (HuggingFace ID)	Ground Truth	Ours Score	Ours AUC	TRAP Score	TRAP AUC	ProFLingo Score	ProFLingo AUC	ZeroPrint Score	ZeroPrint AUC
Source	`meta-llama/Meta-Llama-3-8B`	Positive	0.650	**1.000**	0.990	**1.000**	0.520	**1.000**	–	–
SFT / task FT	`allenai/llama-3-tulu-2-8b`	Positive	0.400	**1.000**	0.500	0.889	0.060	0.444	0.280	0.875
SFT / task FT	`dphn/dolphin-2.9.1-llama-3-8b`	Positive	0.450	**1.000**	1.000	**1.000**	0.140	0.889	0.140	0.375
SFT / task FT	`Groq/Llama-3-Groq-8B-Tool-Use`	Positive	0.550	**1.000**	0.090	0.333	0.140	0.889	0.180	0.562
SFT / task FT	`Magpie-Align/Llama-3-8B-Magpie-Align-SFT-v0.3`	Positive	0.450	**1.000**	0.640	**1.000**	0.220	**1.000**	0.460	**1.000**
SFT / task FT	`openchat/openchat-3.6-8b-20240522`	Positive	0.450	**1.000**	0.200	0.444	0.160	0.944	0.390	**1.000**
Preference / RL	`dfurman/Llama-3-8B-Orpo-v0.1`	Positive	0.650	**1.000**	0.720	**1.000**	0.340	**1.000**	0.430	**1.000**
Language / Domain	`aaditya/Llama3-OpenBioLLM-8B`	Positive	0.400	**1.000**	0.540	0.889	0.180	**1.000**	0.240	0.750
Language / Domain	`DeepMount00/Llama-3-8b-Ita`	Positive	0.300	0.444	0.100	0.333	0.140	**0.889**	0.240	0.750
Language / Domain	`hfl/llama-3-chinese-8b`	Positive	0.550	**1.000**	0.700	**1.000**	0.260	**1.000**	0.300	0.875
Language / Domain	`winninghealth/WiNGPT2-Llama-3-8B-Chat`	Positive	0.400	**1.000**	0.150	0.333	0.020	0.056	0.240	0.750
Pruning	`RedHatAI/SparseLlama-3-8B-pruned_50.2of4`	Positive	0.350	0.778	0.890	**1.000**	0.160	0.944	0.130	0.312
Merging	`mlabonne/Llama-3-SLERP-8B`	Positive	0.400	**1.000**	0.750	**1.000**	0.380	**1.000**	0.460	**1.000**
Merging	`Weyaxi/Einstein-v6.1-Llama3-8B`	Positive	0.550	**1.000**	0.430	0.889	0.120	0.833	0.290	0.875
Independent	`01-ai/Yi-6B`	Negative	0.150	–	0.330	–	0.120	–	0.180	–
Independent	`EleutherAI/llemma_7b`	Negative	0.350	–	0.200	–	0.060	–	0.170	–
Independent	`Qwen/Qwen2.5-7B`	Negative	0.200	–	0.250	–	0.060	–	0.130	–
Independent	`allenai/tulu-2-dpo-7b`	Negative	0.350	–	0.000	–	0.080	–	0.210	–
Independent	`lmsys/vicuna-13b-v1.3`	Negative	0.350	–	0.560	–	0.020	–	0.000	–
Independent	`lmsys/vicuna-7b-v1.5`	Negative	0.300	–	0.000	–	0.160	–	0.270	–
Independent	`meta-llama/Llama-2-13b-chat-hf`	Negative	0.250	–	0.000	–	0.040	–	0.020	–
Independent	`mistralai/Mistral-7B-v0.3`	Negative	0.350	–	0.200	–	0.060	–	0.330	–
Independent	`THUDM/chatglm3-6b-base`	Negative	0.300	–	0.250	–	0.060	–	–	–

are brittle under these Mistral transformations, while ProFLingo and ZeroPrint improve to 0.892 and 0.918. Ours remains at AUC 1.000; the independent average score remains lower despite a comparable independent TestAcc of 0.648.

Cases where Ours is not the row-wise best. There are two source/family rows in the per-suspect table where the row-level AUC of Ours is below another method. Both occur in the Llama3 family and both are narrow subgroup effects rather than all-family failures.

The first exception is `DeepMount00/Llama-3-8b-Ita`. The project lineage file marks it as a strict first-level derivative with an exact `meta-llama/Meta-Llama-3-8B` base-model tag and describes it as an Italian SFT candidate with full 8B safetensors. Its MMLU diagnostic TestAcc is 0.660, so the exception is not caused by a low-quality or nonfunctional model. Instead, the Italian SFT changes the model's answer-format and label-selection behavior enough that the source-optimized hard-label fingerprint is only partially preserved: Ours scores 0.300 on this derivative. That score is moderate, but it overlaps the upper tail of the Llama3 independent negatives, where several unrelated models also score 0.300–0.350 under the same prompt template. The overlap lowers the row-level AUC to 0.444. ProFLingo is better on this single row because its independent background scores are lower in this comparison, not because the derivative has poor MMLU ability. We therefore interpret this as a Llama-specific transfer/separation issue caused by domain SFT and answer-format sensitivity, rather than as a failure of the clean MMLU task.

The second exception is `RedHatAI/SparseLlama-3-8B-pruned_50.2of4`. The lineage audit identifies it as a strict first-level derivative of `meta-llama/Meta-Llama-3-8B` with an exact HF base-model finetune tag; the model name and local recommendation indicate a 50.2of4 structured-sparsity pruning design. Its MMLU diagnostic TestAcc is 0.650, again above the gate and close to other Llama derivatives, so the model is still a valid task-performing suspect. The weakness is specific to the fingerprint signal: Ours scores 0.350 on the pruned checkpoint, which ties the highest independent Ours scores in the Llama3 pool. Because row-level AUC compares this single derivative against all independent negatives, those ties reduce the row-level

TABLE 25: Per-suspect comparison using **Mistral-7B-v0.3** as the protected source model. Family-derived suspects are shown in the upper block and independent suspects in the lower block. Score denotes FSR or method-specific fingerprint score, and AUC denotes family-versus-independent separability.

Category	Suspect Model (HuggingFace ID)	Ground Truth	Ours Score	Ours AUC	TRAP Score	TRAP AUC	ProFLingo Score	ProFLingo AUC	ZeroPrint Score	ZeroPrint AUC
Source	mistralai/Mistral-7B-v0.3	Positive	0.600	**1.000**	0.250	**1.000**	0.660	**1.000**	0.570	**1.000**
SFT / task FT	amdevraj/mistral-7b-ift	Positive	0.600	**1.000**	0.000	0.500	0.220	**1.000**	0.310	**1.000**
SFT / task FT	cypienai/cymist-2-v03-SFT	Positive	0.400	**1.000**	0.000	0.500	0.260	**1.000**	0.310	**1.000**
SFT / task FT	migtissera/Tess-3-7B-SFT	Positive	0.450	**1.000**	0.000	0.500	0.200	**1.000**	0.320	**1.000**
Preference / RL	lamm-mit/mistral-7B-v0.3-Base-CPT-SFT-DPO-09022024	Positive	0.300	**1.000**	0.000	0.500	0.160	**1.000**	0.320	**1.000**
Preference / RL	llmat/Mistral-v0.3-7B-ORPO	Positive	0.350	**1.000**	0.000	0.500	0.200	**1.000**	0.470	**1.000**
Language / Domain	BEE-spoke-data/Mistral-7B-v0.3-stepbasin-books-20k	Positive	0.600	**1.000**	0.170	**1.000**	0.500	**1.000**	0.580	**1.000**
Language / Domain	entfane/math-virtuoso-7B	Positive	0.450	**1.000**	0.000	0.500	0.180	**1.000**	0.270	0.800
Language / Domain	klcsp/mistral7b-fft-coding-11-v1	Positive	0.350	**1.000**	0.000	0.500	0.000	0.000	0.170	0.200
Language / Domain	openfoodfacts/spellcheck-mistral-7b	Positive	0.550	**1.000**	0.000	0.500	0.380	**1.000**	0.590	**1.000**
Language / Domain	pszemraj/Mistral-7B-v0.3-sarcasm-scrolls-4k	Positive	0.450	**1.000**	0.000	0.500	0.120	**1.000**	0.390	**1.000**
Language / Domain	pucpr-br/Clinical-BR-Mistral-7B-v0.2	Positive	0.250	**1.000**	0.000	0.500	0.060	0.250	0.350	**1.000**
Language / Domain	silversword/US-Immigration-Law-Mistral-7B-v0.3-CPT	Positive	0.550	**1.000**	0.000	0.500	0.440	**1.000**	0.570	**1.000**
Pruning	ai-and-society/mistral-7B-Instruct-v0.3-wanda-wanda-unstruct-50	Positive	0.350	**1.000**	0.000	0.500	0.400	**1.000**	0.300	0.900
Pruning	IntelLabs/sqft-mistral-7b-v0.3-50-base	Positive	0.400	**1.000**	0.000	0.500	0.140	**1.000**	0.360	**1.000**
Merging	mlfoundations-dev/hp_ablations_grid_mistral_base-mistralv0.3	Positive	0.350	**1.000**	0.000	0.500	0.100	0.917	0.260	0.800
Merging	mlfoundations-dev/mistral_7b_0-3_oh-dcft-v3.1-llama-3.1-405b	Positive	0.450	**1.000**	0.000	0.500	0.400	**1.000**	0.270	0.800
Independent	01-ai/Yi-6B	Negative	0.150	–	0.000	–	0.060	–	0.250	–
Independent	EleutherAI/llemma_7b	Negative	0.200	–	0.000	–	0.080	–	0.230	–
Independent	Qwen/Qwen2.5-7B	Negative	0.200	–	0.000	–	0.060	–	0.120	–
Independent	meta-llama/Meta-Llama-3-8B	Negative	0.100	–	0.000	–	0.080	–	0.300	–
Independent	microsoft/Phi-3-small-8k-instruct	Negative	0.100	–	0.000	–	0.100	–	0.190	–
Independent	THUDM/chatglm3-6b-base	Negative	0.200	–	0.000	–	0.060	–	–	–

AUC to 0.778. A plausible mechanism is that structured pruning preserves much of the clean multiple-choice capability while perturbing the internal target-answer preference induced by the GCG prefix; in other words, task accuracy is preserved better than fingerprint logit/label preference. TRAP, by contrast, uses retrieval-style artifacts and happens to remain high on this pruning checkpoint, giving it AUC 1.000 in this one-row subgroup. Since the pruning panel contains only one family model plus the source, it is particularly sensitive to this single derivative; the all-family Llama3 result still favors Ours (AUC 0.944).

ChatGLM and ZeroPrint handling. We include `THUDM/chatglm3-6b-base` as an independent negative control for Ours, TRAP, and ProFLingo whenever valid hard-label artifacts are available. Its ZeroPrint entry is shown as "–" because the official/repo-aligned ZeroPrint generation pipeline encountered a third-party remote-code compatibility issue for ChatGLM under the Transformers version used in our run. ZeroPrint AUCs are therefore computed over the independent models with valid ZeroPrint artifacts. The Mistral-group `microsoft/Phi-3-small-8k-instruct` ZeroPrint entry was recomputed with a compatibility patch and incorporated into the cached source-row artifact. These compatibility notes affect only independent negative controls and do not change any source/family positive row.

9. Additional Naturalness and Robustness Diagnostics

9.1. Prompt-Embedding Diagnostics. The main text includes a prompt-embedding t-SNE figure to qualitatively inspect whether naturalized prompts move closer to natural prompt regions while retaining verification performance. The visualization uses a fixed external sentence-embedding model only for analysis. It is not used during fingerprint construction, target selection, or black-box verification.

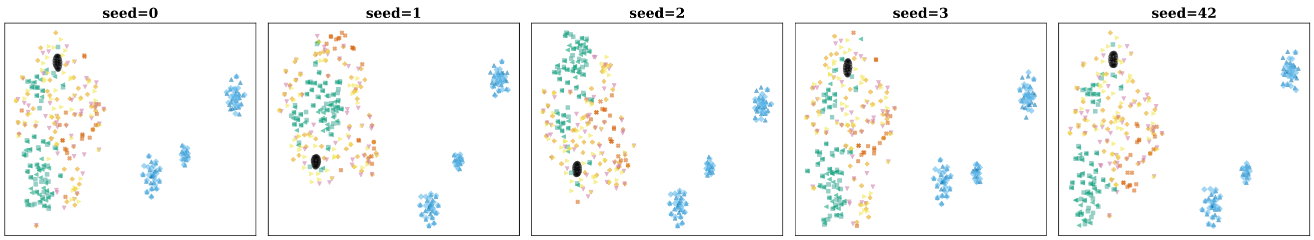

Figure 18: Seed-stability diagnostic for the prompt-embedding t-SNE visualization.

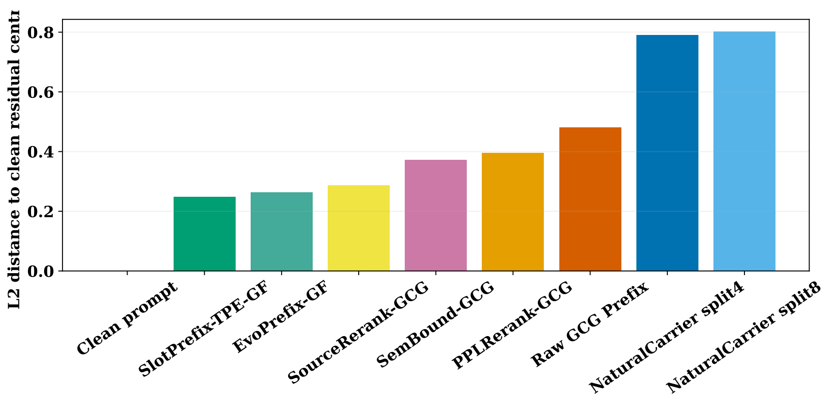

Figure 19: Embedding-distance diagnostics for naturalized prompt variants. Lower distance indicates closer proximity to the chosen clean or natural prompt reference under the external embedding model.

Observation. Figures 18 and 19 complement the prompt-embedding visualization. The seed-stability plot checks that the qualitative layout is not an artifact of one random seed. The distance-bar plot summarizes how naturalized variants compare with clean or natural references in embedding space. These diagnostics are qualitative only; the actual verifier still uses the hard-label target-hit score.

9.2. Distillation Step Sweep. We also evaluate a smaller clean-query distillation setting with 250 teacher-label pairs. Table 26 shows that the student becomes more similar to the teacher on clean questions, and our verification AUC also increases. This suggests that the fingerprint signal is inherited as the student absorbs teacher behavior, even though the private fingerprint prompts are not used for distillation.

TABLE 26: Step sweep for clean-query model distillation from Mistral-7B-v0.3 to Llama-2-7B using 250 clean teacher-label pairs. Clean100 is measured on the 100-example MMLU-CF subset [42], TA denotes teacher agreement, and fingerprint rows report student-vs-independent AUC.

Metric	S0	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
Clean100	0.66	0.65	0.69	0.75	0.73	0.77	0.73	0.71	0.71	0.72	0.72
TA	0.68	0.64	0.67	0.73	0.73	0.75	0.73	0.73	0.71	0.72	0.72
Ours	0.31	0.56	0.75	0.56	0.75	0.88	0.81	0.88	0.88	0.88	0.88
TRAP	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
ProF.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.06	0.00	0.00	0.00
ZPrint	0.61	0.22	0.22	0.11	0.11	0.00	0.11	0.11	0.11	0.11	0.11